

Practice Problems:
AC Power
Introduction to Electric Circuits
Supplemental Instruction

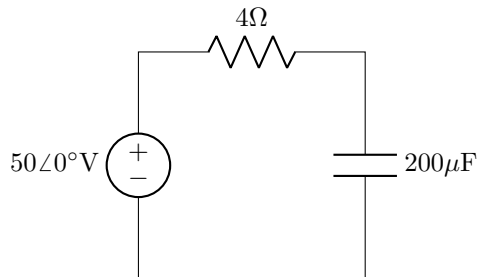
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1 Complex Power in an RC Circuit

Find the average power, reactive power, apparent power, and power factor. Assume $f = 120$ Hz.



Solution

The total impedance is $\mathbf{Z} = 4 + \frac{1}{j \cdot 240\pi \cdot 200\mu\text{F}} = 4 - 6.63j = 7.74\angle -58.9^\circ \Omega$. Therefore, the current is $\mathbf{I} = \frac{50\text{V}}{7.74\angle -58.9^\circ \Omega} = 6.456\angle 58.9^\circ \text{ A}$.

Therefore, the average power is

$$P = \frac{V_m I_m}{2} \cos(\alpha_v - \alpha_i) = \frac{50 \cdot 6.456}{2} \cos(0^\circ - 58.9^\circ) = 83.37 \text{ W}$$

The reactive power is

$$Q = \frac{V_m I_m}{2} \sin(\alpha_v - \alpha_i) = \frac{50 \cdot 6.456}{2} \sin(0^\circ - 58.9^\circ) = -138.2 \text{ VAR}$$

The apparent power is

$$S = \frac{V_m I_m}{2} = \frac{50 \cdot 6.456}{2} = 161.4 \text{ VA}$$

The power factor is

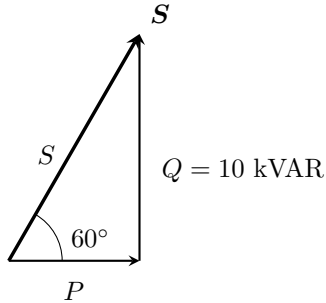
$$PF = \cos(\alpha_v - \alpha_i) = \cos(0^\circ - 58.9^\circ) = 0.517, \text{ leading}$$

2 Power Triangle

Given $Q = 10$ kVAR and a power factor of 0.5, find the average power.

Solution

$\arccos(0.5) = 60^\circ$, so the power triangle is as follows:



We can find average power using the equation

$$\tan(60^\circ) = \frac{10 \text{ kVAR}}{P}$$

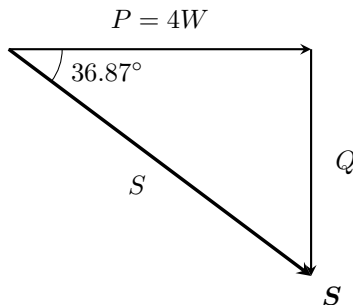
Solving for P , we get $P = 5.77 \text{ kW}$.

3 Power Triangle

Given $P = 4 \text{ W}$ and a power factor of 0.8 leading, draw the power triangle and find S and Q .

Solution

$\arccos(0.8) = 36.87^\circ$, but since current leads, $\alpha_i > \alpha_v \Rightarrow (\alpha_v - \alpha_i) < 0$, so the reactive power is negative:



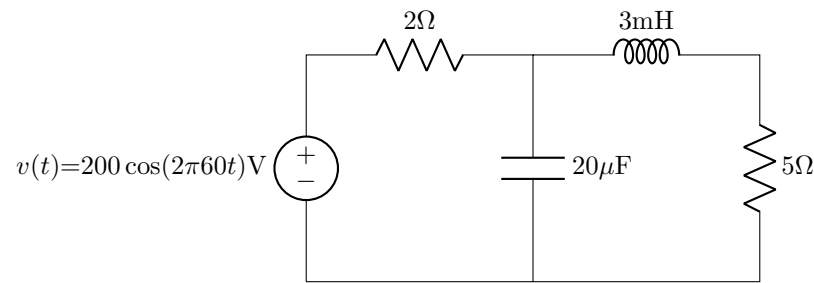
The apparent power can be found using the equation

$$\cos(36.87^\circ) = \frac{4 \text{ W}}{S} \Rightarrow S = 5 \text{ VA}$$

The reactive power could be found using the pythagorean theorem: $Q = -3 \text{ VAR}$.

Complex Power in RLC Circuit

Find the average power, reactive power, apparent power, and power factor.



Solution

The total impedance is

$$\mathbf{Z} = 2 + \frac{1}{j \cdot 120\pi \cdot 20\mu\text{F} + \frac{1}{j \cdot 120\pi \cdot 3\text{mH} + 5}} = 7.079 + 0.948j \Omega$$

The voltage is $\mathbf{V} = 200\angle 0^\circ \text{ V}$, and the current is $\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} = 27.76 - 3.72j = 28\angle -7.62^\circ \text{ A}$. The power factor is

$$PF = \cos(0^\circ - (-7.62^\circ)) = 0.9912, \text{ lagging}$$

The apparent power is

$$S = \frac{200 \cdot 28}{2} = 2.8 \text{ kVA}$$

The average power is

$$P = S \cdot PF = 2.8\text{kVA} \cdot 0.9912 = 2.775 \text{ kW}$$

The reactive power is

$$Q = S \cdot \sin(7.62^\circ) = 371 \text{ VAR}$$

References

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